Hotel Booking and Cancellation Insights

Business Analyst Candidate

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Executive Summary

This report details the comprehensive analysis performed on hotel booking data to identify drivers of cancellations and provide actionable strategies to optimize profitability. Our analysis reveals that high-risk channels and incentive structures are significantly impacting revenue, necessitating a shift toward commitment-based incentives. All technical code and detailed visualizations associated with this study are hosted in the project repository: Hotel Booking Analysis GitHub Repository.

1. Observed Patterns and Trends

Based on our exploratory data analysis, the following trends were identified:

- **Channel Variance:** Booking volume is highly skewed across channels. Certain third-party channels demonstrate significantly higher cancellation rates compared to direct bookings, indicating a lower quality of lead generation.
- **Seasonality of Demand:** Booking volume and stay lengths fluctuate significantly based on the month, confirming the influence of seasonal travel patterns on our inventory occupancy.
- **Star Rating Disparities:** Cancellation behavior is not uniform; mid-to-high tier properties show distinct volatility patterns, likely due to a more price-sensitive demographic seeking alternative options post-booking.

2. Root Cause Explanations

To address the "Why" behind these trends, we analyzed key behavioral indicators:

- **High Lead Time Correlation:** We observed a strong correlation between long lead times (the gap between booking and travel) and cancellation rates. Customers booking far in advance exhibit lower commitment, frequently treating reservations as "soft holds."
- **Incentive-Driven Cancellations:** Reliance on broad-based couponing and cash-back schemes has attracted price-sensitive customers who lack brand loyalty, making them more likely to cancel if a cheaper deal emerges.
- **Operational Friction:** The lack of tiered cancellation terms has inadvertently encouraged booking-and-comparing behavior, as users bear no financial risk when canceling.

3. Actionable Business Recommendations

Based on our findings, we propose the following strategic shifts:

- **Tiered Cancellation Strategy:** Introduce a two-tier pricing model: *Flexible* (higher price, free cancellation) and *Non-Refundable* (discounted, no cancellation). This reduces inventory loss by pricing the risk of cancellation.
- **Loyalty-Based Rewards:** Transition from immediate coupon redemptions to a loyalty program that rewards guests only after the *completion* of their stay. This will improve repeat booking rates and lifetime value.
- **Channel Rationalization:** Shift marketing spend from channels identified as high-risk/low-margin toward channels that demonstrate higher guest retention and lower cancellation rates.
- **Dynamic Pricing Implementation:** Utilize seasonal demand trends to implement demand-based pricing, ensuring maximum occupancy during off-peak periods without compromising the Average Daily Rate (ADR).

Appendix: Technical Methodology

The analysis was conducted using Python, utilizing `pandas` for data manipulation and `seaborn/matplotlib` for visualization.

Data Cleaning Process

The raw dataset was cleaned to ensure accurate results:

1. **Date Conversion:** All date-related columns were cast to datetime objects to facilitate lead-time calculations.
2. **Feature Engineering:** New features were created, including 'lead_time' (gap between booking and travel), 'is_canceled' (binary flag), and 'net_profit' (selling price minus costs and cashback).
3. **Outlier Handling:** Records with negative stay lengths or zero values were excluded to maintain model integrity.

Repository Access

All code, including cleaning scripts, statistical analysis, and visual generation, can be reviewed in the repository: `https://github.com/komal-ti/Hotel_booking_analysis`. This repository serves as the definitive record of the project methodology.